



Selected topics in Cryptography

Quiz 2B

March 5, 2026

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Problems

Read the following problem and solve it. Please write down the detail of all your calculations.

1. Answer the following questions about Algorithm 2 :

- When does the algorithm do the maximum number of operations i.e. point addition and doubling? Why? (Correct answer: 1 point)
- Use it to compute $38P$. Write down the values of k in binary, and the values of i, k_i, Q, P for each iteration. Fill the table below. (Correct answer: 4 points)

Algoritmo 2: Right-to-left binary method for point multiplication(k, P)

Input : $k = (k_{t-1}, \dots, k_1, k_0)_2$ binary representation of k ; $P \in \mathbb{E}(\mathbb{Z}_p)$

Output: kP

```
1  $Q \leftarrow \infty$  ;
2 for  $i \leftarrow 0$  to  $t-1$  do
3   if  $k_i == 1$  then
4      $Q \leftarrow Q + P$ ;
5   end
6    $P \leftarrow 2P$ 
7 end
8 return  $Q$  ;
```

i	k_i	Q	P
0	0	∞	$2P$
1	1	$2P$	$4P$
2	1	$2P + 4P = 6P$	$8P$
3	0	$6P$	$16P$
4	0	$6P$	$32P$
5	1	$6P + 32P = 38P$	$64P$

- (a) Cuando la representación binaria de k está compuesta de únicamente unos. Porque siempre entrara en la condición del `if` en cada una de sus k ; por lo que realizara tanto la suma de $P + 0$ como el doblado $2P$.

(b) $38P$

1	0	0	1	1	0
32	16	8	4	2	1
k_5	k_4	k_3	k_2	k_1	k_0